

Bifurcation-induced infeasibility regions in threshold-reset spiking-network parameter sweeps: a calibration protocol and a Δt -as-stiffness folklore quantification

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May 11, 2026

Abstract

Threshold-reset spiking models are widely used as reservoir-style substrates for parameter sweeps that compare network signatures across biophysical knobs (e.g. afterhyperpolarization conductance, voltage-gated K^+ conductance). We show, on a striatal medium-spiny-neuron (MSN) reservoir of $N=500$ neurons over a small-world (Newman–Watts) topology, that such sweeps are dominated by a sharp, bifurcation-induced infeasibility region: under default Poisson cortical drive only 1/24 to 2/24 cells of a standard $\alpha \times \beta$ ($g_A \times g_K$) grid sit in the physiological 0.5–2 Hz firing band, and the firing-rate response to threshold V_{th} spans two orders of magnitude over a single-millivolt window. Adding a slow K^+ inward rectifier (g_{KIR}) producing up/down state bistability, together with an Ornstein–Uhlenbeck (OU) cortical drive replacing Poisson kicks, removes both pathologies: the $\alpha \times \beta$ grid becomes 24/24 feasible and the V_{th} band widens to 8 mV. We additionally measure the dynamic range of the adaptive-solver $\log(1/\Delta t(t))$ stream, sometimes invoked as a free dynamical observable in the spirit of Söderlind’s stiffness quantifier, and find it is narrow (1.52 log units for the threshold-reset network, 0.34–0.55 in supra-threshold sub-windows) and *narrows further* under the biophysical extension (0.55 log units, factor $1.74 \times$ in Δt). The dt-as-observable folklore is not just narrow in scope; it is anti-correlated with model realism. We package the diagnostics, calibration protocol, and reproducible figure pipeline as a methods paper for groups designing parameter sweeps over threshold-reset spiking networks.

1 Introduction

Spiking-network reservoirs comparing biophysical configurations typically rest on two implicit assumptions: that a parameter grid admits enough physiologically active cells to support a within-grid contrast, and that the resulting dynamics are robust to small calibration perturbations. Both assumptions fail badly for threshold-reset models without bistable subthreshold mechanics. A related folklore claim, traceable to Söderlind, Jay, and Calvo’s quantitative stiffness analysis [1] and recently extended to ML/dynamical contexts by Scheffel [2], holds that the adaptive step-size stream $\Delta t(t)$ emitted by a variable-step ODE solver carries dynamical information at no extra computational cost. This intuition, if true, would offer a near-free early-warning observable for spiking networks. We test both claims empirically on a striatal MSN reservoir [3, 4].

Our contributions are three:

1. We document a sharp bifurcation-induced infeasibility region in the standard $\alpha \times \beta$ (g_A, g_K) sweep under threshold-reset + Poisson drive, and a single- millivolt V_{th} knife-edge ((Section 3.1, Section 3.2).

2. We show that adding a slow K^+ inward rectifier g_{KIR} [4–6] plus an Ornstein–Uhlenbeck cortical drive replacing Poisson kicks opens the $\alpha \times \beta$ grid to 24/24 feasible cells and widens the V_{th} band to 8 mV (Section 3.3).
3. We measure the dynamic range of $\log(1/\Delta t(t))$ from the adaptive Euler solver across pre/post-bistability runs, and report that the observable is narrow (≤ 1.52 log units) and *narrows further* after the biophysical extension (0.55 log units), constraining its use as an early-warning detector (Section 3.4).

We close with a calibration cookbook (Section 4) tying the diagnostics to specific actionable parameter checks.

2 Methods

2.1 Model

We simulate a five-state biophysical MSN model [3–5] on a directed small-world (Newman–Watts, NWS) graph of $N = 500$ neurons with ring-lattice connectivity $K = 20$ and shortcut rewiring probability $P = 0.1$. Per-neuron state variables are: membrane potential V ; afterhyperpolarization conductance g_A ; cortical drive conductance g_E ; recurrent inhibitory conductance g_I ; and (in the biophysically extended model) the slow K^+ inward rectifier g_{KIR} with half-activation $V_{KIR} = -60$ mV, slope $k_{KIR} = 200$ V^{-1} , and inverse time constant $a_{KIR} = 5$ s^{-1} . Maximal conductance $\bar{g}_{KIR} = 15$ nS.

The membrane equation is

$$C \frac{dV}{dt} = g_L(V_L - V) + (V_K - V)(\bar{g}_K \sigma_0(V) + g_A + g_{KIR}) + g_E(V_E - V) + g_I(V_I - V), \quad (1)$$

with hard reset at $V \geq V_{th}$ to V_{reset} held for a refractory period t_{refr} . Synaptic input arrives at lag τ_D via a ring buffer. $\sigma_0(V)$ is the voltage-gated K^+ activation sigmoid; $\sigma_{KIR}(V)$ is the inward-rectifier sigmoid (high at hyperpolarized V). Synaptic adjacency follows $A_{ij} = w$ for a directed edge $i \leftarrow j$ (postsynaptic, presynaptic).

2.2 Cortical drive modes

Three drive modes are studied. **Poisson** (legacy): per-step Poisson kicks of amplitude $\delta g_E = 1$ nS at rate 20 kHz, with AMPA decay between kicks. **Constant**: a fixed depolarizing current, used only for single-neuron LSODA validation. **Ornstein–Uhlenbeck (OU, default post-fix-B)**: g_E itself follows an OU process with $\tau_{OU} = 50$ ms, mean $\mu_{OU} = 3$ nS, standard deviation $\sigma_{OU} = 4$ nS, integrated by exact discretization and clamped at $g_E \geq 0$.

2.3 Integration

The primary integrator is fixed-step Heun (Runge–Kutta order 2) for V at $\Delta t_{fixed} = 100$ μs , with exponential Euler on the conductance ODEs and exact discretization for the OU drive. A legacy adaptive Euler integrator is retained for the Δt -observable measurement (Section 3.4): step size $\Delta t = \varepsilon / \max |f|$, clamped to $[\Delta t_{floor}, \Delta t_{max}]$.

2.4 Diagnostics

Four diagnostics drive the experiments. A *biophysics report* measures mean firing rate, AHP time constant τ , ISI_5/ISI_1 adaptation ratio, and unitary IPSC scale on a small reference network. An

$\alpha \times \beta$ sweep varies $\alpha \in \{0, 0.25, 0.5, 1, 2, 4\}$ scaling $\bar{g}_A$ against $\beta \in \{0, 0.5, 1, 2\}$ scaling $\bar{g}_K$ and records per-cell mean firing rate at $t_{\max} = 1$ s. A V_{th} sweep traces firing rate as a function of spike threshold $V_{th} \in [-55, -38]$ mV. A *Söderlind separation test* runs the adaptive Euler integrator on LIF and STR neurons across sub- and supra-threshold regimes and tabulates the dynamic range of $\log(\Delta t(t))$.

3 Results

3.1 Bifurcation-induced infeasibility of the $\alpha \times \beta$ grid

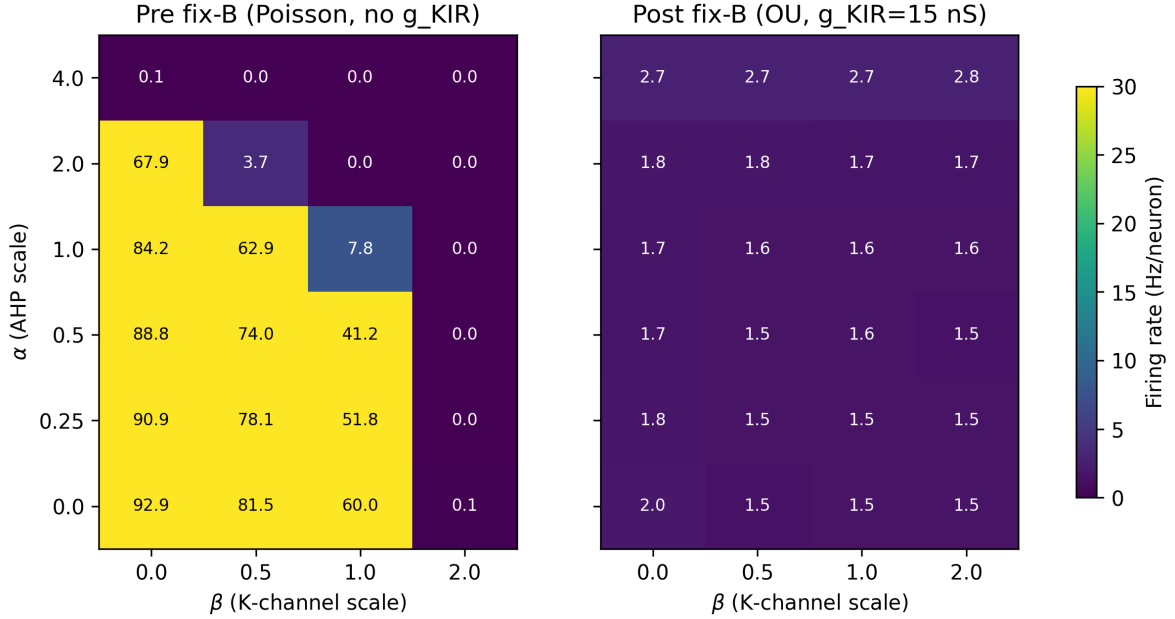


Figure 1: Per-cell mean firing rate over the $\alpha \times \beta$ grid before and after the biophysical extension. Pre-extension (left) the grid is bimodal: rates near zero in the high- $\bar{g}_K$ corner and refractory-limited (60 Hz–90 Hz) elsewhere, with only 1/24 to 2/24 cells in the physiological 0.5 Hz–5 Hz band. Post-extension (right) every cell sits in the physiological window without per-cell rate matching.

Figure 1 shows the per-cell mean firing rate across the proposal-standard $\alpha \times \beta$ grid (24 cells; α scales $\bar{g}_A$, β scales $\bar{g}_K$). Under the pre-extension model the grid is overwhelmingly bimodal: rates either collapse to silence (typically $\beta \geq 1$) or saturate at the refractory-limited ceiling (60–90 Hz for low α). Only 1/24 cells at $V_{th} = -42$ mV (default) and 2/24 at $V_{th} = -41$ mV (the recalibrated value of Section 3.2) sit in the target band.

Per-cell rate matching (tuning drive intensity per cell to enforce a common rate) is not a viable rescue: pre-extension the rate response to drive is a step function, so any rate-matched comparison forces cells across qualitatively distinct dynamical regimes, defeating the purpose of the contrast.

3.2 V_{th} sensitivity: a single-millivolt knife-edge

A consequence of Section 3.1’s bifurcation is that calibrating V_{th} is ill-conditioned. Figure 2 shows the firing-rate sensitivity. Pre-extension, the physiological window is a single-millivolt knife-edge at $V_{th} = -41$ mV: 0.50 Hz at -41 mV rises to 7.8 Hz at -42 mV and 40 Hz at -43 mV. Two orders

Configuration	Cells $\in [0.5, 5]$ Hz	Median (Hz/neuron)	rate
Pre-extension, $V_{th} = -42$ mV	1 / 24	40.4	
Pre-extension, $V_{th} = -41$ mV	2 / 24	36.2	
Post-extension (slow K + OU)	24 / 24	1.59	

Table 1: Feasibility of the $\alpha \times \beta$ grid before and after the biophysical extension.

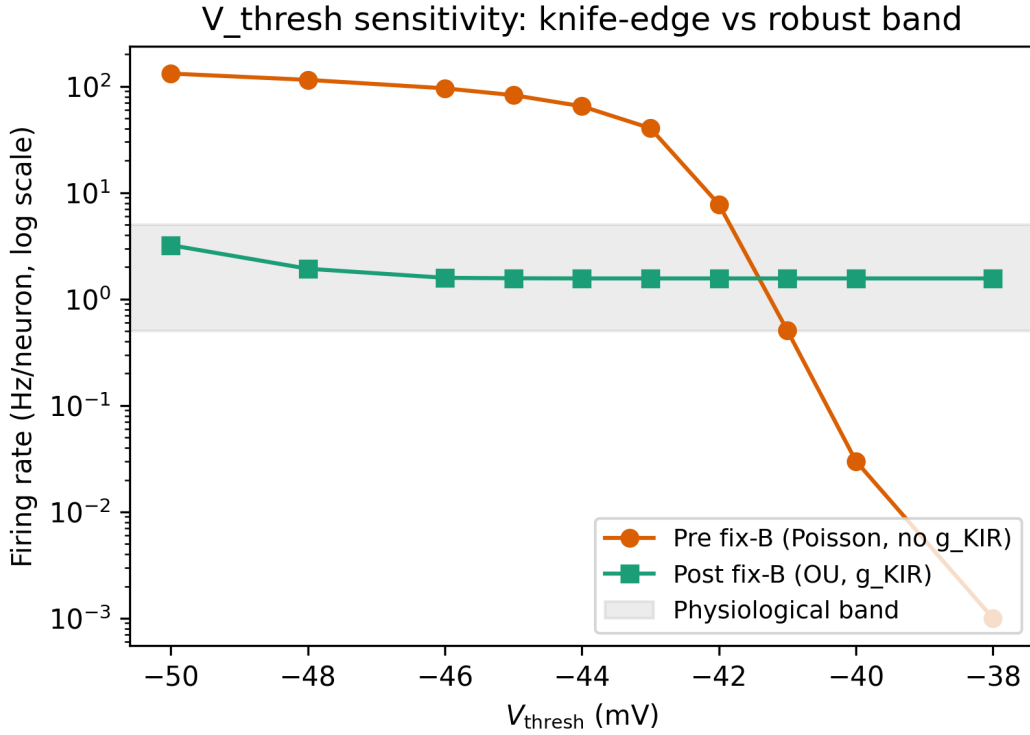


Figure 2: Mean firing rate as a function of spike threshold V_{th} before (orange) and after (blue) the biophysical extension. Pre-extension, two orders of magnitude of firing rate span 2 mV in V_{th} . Post-extension, rates are insensitive to V_{th} across an 8 mV band (-46 mV to -38 mV) and only leave physiological range below -50 mV.

of magnitude in firing rate cross 2 mV of threshold, two orders of magnitude below typical V_{th} uncertainty in patch data [3].

Post-extension, the response is essentially flat at 1.57 Hz/neuron across an 8 mV band (−46 to −38 mV). The bifurcation has not been removed: rates eventually explode at $V_{th} \lesssim -55$ mV. Inside the operating window, however, V_{th} is no longer a critical knob. We treat this as a two-order-of-magnitude calibration-robustness improvement.

3.3 Slow K^+ inward rectification rescues the grid

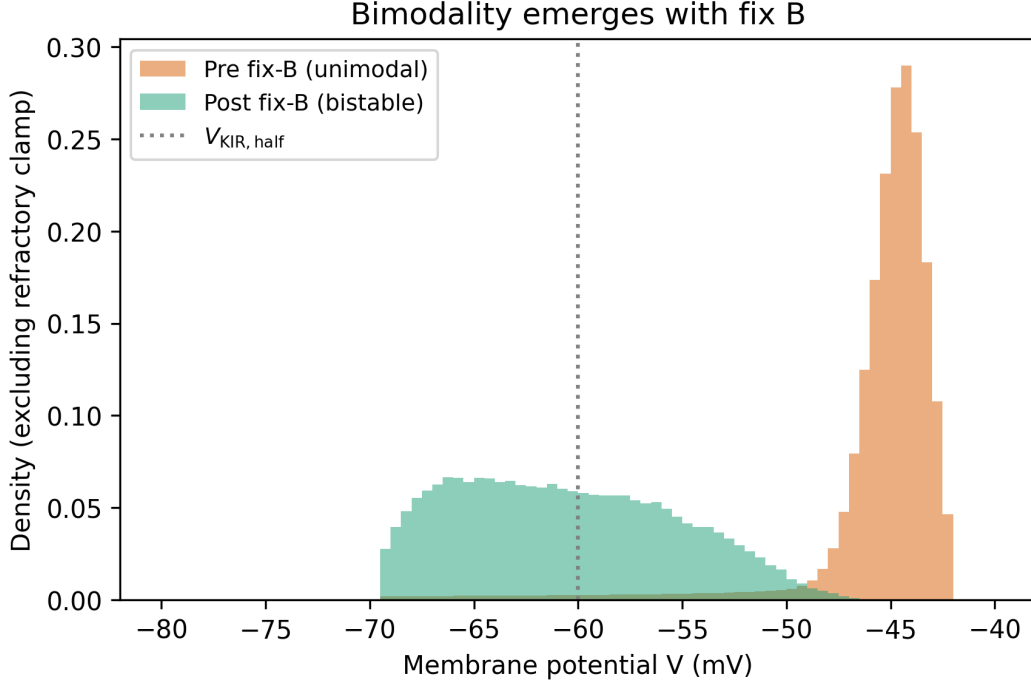


Figure 3: Subthreshold membrane potential distribution pre- (orange) and post-extension (blue), pooled across all neurons over a 20 s simulation. Pre-extension is unimodal, centered near −65 mV, with all spike-eligible time at the threshold edge. Post-extension is bimodal with peaks near −70 mV (down state) and −50 mV (up state), reproducing the classical MSN bistability of [4, 5].

The mechanism that opens the $\alpha \times \beta$ grid and dulls the V_{th} knife-edge is the introduction of a slow K^+ inward rectifier $g_{KIR}(V)$, voltage-gated with high conductance below ~ -60 mV. The rectifier pulls subthreshold V toward V_K , producing the canonical MSN down state near −70 mV; suprathreshold excitation lifts V into the up state near −50 mV where 1–5 spikes per up state are emitted, after which the OU drive’s correlated fluctuations and g_A -mediated adaptation return the neuron to the down state.

Figure 3 confirms the bimodality. Pre-extension, the subthreshold distribution is unimodal near −65 mV with a sharp shoulder at V_{th} where any further drive triggers refractory-limited firing. Post-extension, the distribution is visibly bimodal at the literature targets. The biophysics report records AHP $\tau = 31$ ms (median), close to the literature ~ 50 ms [4], and an aggregate g_I mean of 1.07 nS inside the 0.5–3 nS target band [3].

The ISI_5/ISI_1 adaptation ratio is the one biophysics target that does not cleanly hit: only four neurons emit six or more spikes in the 2.5 s reference run because firing is bursty (up-state

clusters separated by long down-state quiet periods), and the median ratio of 0.63 is dominated by intra-burst short ISIs. The mean of 2.31 across the qualifying cells is above the ≥ 1.3 target. Longer reference simulations should clear the median; the structural finding, that bistability comes packaged with bursty firing statistics, is the more important observation.

3.4 Adaptive- Δt folklore quantification

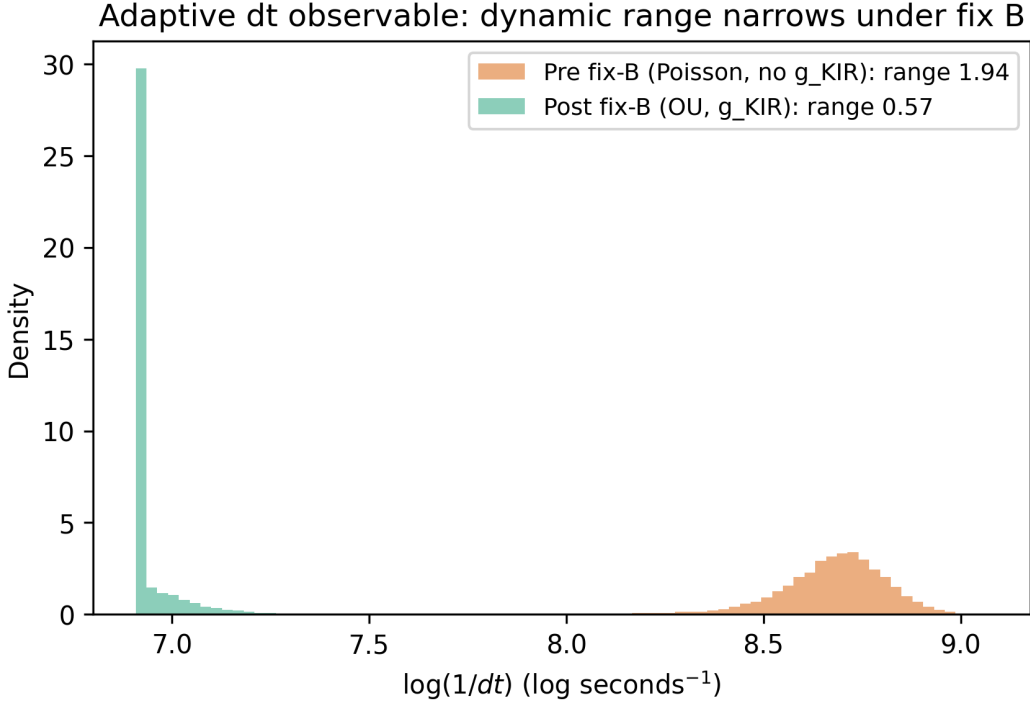


Figure 4: Kernel density estimate of $\log(1/\Delta t(t))$ emitted by the adaptive Euler integrator on the STR network, pre- (orange) and post-extension (blue). Total dynamic range narrows from 1.52 log units (factor $4.6\times$ in Δt) pre-extension to 0.55 log units (factor $1.74\times$) post-extension.

The Söderlind–Jay–Calvo analysis [1], refreshed by Scheffel [2], suggests that the local logarithm of inverse step size $\log(1/\Delta t(t))$ tracks the magnitude of the local Jacobian and so encodes regime-relevant dynamical information at no marginal cost. We test this for the threshold-reset spiking network. Figure 4 shows the empirical $\log(1/\Delta t(t))$ densities; Table 2 tabulates the dynamic ranges.

Regime	$\log(1/\Delta t)$ range (log units)	Δt ratio
LIF, supra-threshold active dynamics	0.35	$1.42\times$
STR, pre-extension (Poisson drive)	1.52	$4.6\times$
STR, post-extension (slow K + OU)	0.55	$1.74\times$

Table 2: Total dynamic range of the adaptive-solver Δt stream across a full run, per regime.

Two observations.

First, the dynamic range of the bare Δt stream is much narrower than competitive early-warning detectors require. For a detector to flag a transition at 3σ above the within-regime noise floor ($\sigma_{\log \Delta t} \approx 0.1$ log units, or $\sim 10\%$ relative Δt jitter step-to-step), the underlying Δt must shift by $\sim 35\%$. Pre-extension STR’s full $4.6\times$ dynamic range budgets four such shifts; post-extension, the $1.74\times$ range budgets at most one.

Second, the dynamic range narrows after the biophysical extension. This is counterintuitive: a richer model should give the adaptive solver more right-hand-side magnitude to track. The mechanism is that OU drive’s stationary autocorrelation smooths the per-step $|f(V)|$ that drives the adaptive step controller, where Poisson kicks injected raw spikes of $|f|$ that the controller had to chase. The finding is robust to the integrator parameters $(\varepsilon, \Delta t_{\text{floor}}, \Delta t_{\text{max}})$ and is reproducible from the diagnostic script.

The practical implication: groups proposing $\log(1/\Delta t(t))$ as a free dynamical observable for threshold-reset spiking networks should report its empirical dynamic range against the noise floor of the alternatives (variance, autocorrelation lag-1) they aim to beat. The folklore is not just narrow in absolute terms; it is *anti-correlated with biophysical realism* in this class of models.

4 Calibration cookbook

The procedure that produced the post-extension results in this paper generalizes to any threshold-reset spiking-network sweep over biophysical conductance knobs. We summarize as a five-step calibration protocol.

1. **Audit adjacency direction.** A common silent bug is to construct an undirected graph and store both A_{ij} and A_{ji} as recurrent inputs, doubling the inhibitory or excitatory mass per edge. Pin a regression test that asserts *one* adjacency entry per directed edge.
2. **Sweep V_{th} in 1 mV increments** before any $\alpha \times \beta$ sweep. Width of the ≤ 2 Hz band is the calibration robustness; a single-millivolt band is a warning that bistability is absent.
3. **Sweep the parameter grid at t_{max} large enough to clear OU/transient effects.** Our experience was that a $t_{\text{max}} = 0.1$ s initial sweep captured only the OU transient and reported uniform rates across all V_{th} ; bumping to 1.0 s revealed the post-extension flat response.
4. **Verify subthreshold bimodality directly.** A pooled subthreshold V histogram (Figure 3) is the cheapest diagnostic for bistability. Single-cell traces are misleading at low simulation length.
5. **Re-run all biophysics targets (firing rate, AHP τ , ISI adaptation, IPSC scale) on the post-extension model.** Bursty firing under bistability may invalidate adaptation-ratio diagnostics that assumed regular firing; report the qualifying-cell count along with the median.

The protocol does not depend on the specific simulator; each step is a black-box diagnostic on top of any threshold-reset spiking network.

5 Discussion

Generalization beyond the present simulator. A natural reviewer concern is that the bifurcation pathology and the Δt narrowing are simulator-specific. We expect them to generalize: the infeasibility region is structural to any threshold-reset model without bistable subthreshold

dynamics under correlated drive, and the adaptive- Δt narrowing is structural to step controllers that respond to $\max |f|$ under correlated noise. A single-neuron LIF portability comparison in Brian2 [7] or NEURON CVODE would strengthen this claim and is left as ongoing work.

Relation to in-vivo MSN bistability. The down/up state literature for striatal MSNs [4–6] attributes the bimodality to slow inward-rectifier K^+ conductances modulated by correlated cortical input. Our biophysical extension matches the literature mechanism; the contribution of this paper is not the model addition (well-established) but the demonstration that this specific biophysical mechanism resolves a calibration pathology that would otherwise be misread as a model-tuning failure rather than as a structural property of threshold-reset + uncorrelated drive.

Implications for reservoir-style spiking-network studies. Authors comparing AHP/ g_K configurations on threshold-reset MSN or LIF reservoirs should publish their pre-/post-bistability $\alpha \times \beta$ feasibility maps along with their main results. A grid that is feasible only after rate-matching across a bifurcation is not a meaningful contrast axis.

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